iLearn Practice Exam

10BR · Chapter / Topic 4

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 4: Software

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 13 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

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1 Complete the statements about different types of software.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

application assembly language bootloader central processing unit (CPU)
firmware hardware operating output system user

..... software provides the services that the computer requires; an example is utility software.

..... software is run on the operating system.

The ..... system is run on the firmware, which is run on the ..... .

[4]

2 A user has both system software and application software installed on their computer.

(a) Describe the difference between system software and application software.

Give an example of each software in your answer.

.....

.....

.....

.....

.....

.....

..... [4]

(b) State which component in the computer would store both types of software when the power is turned off.

..... [1]

3 A programmer writes a computer program in a high-level language.

(a) Tick (✓) **one** box to show which statement is a benefit of writing a program in a high-level language, instead of a low-level language.

A The program can directly manipulate the hardware. ☐

B The program is machine independent. ☐

C The program is more memory efficient. ☐

D The program is quicker to execute. ☐

[1]

(b) Translators are used to translate the high-level language so that it can be processed by the computer.

(i) State what the high-level language is translated into.

..... [1]

(ii) One translator converts and executes the code line by line.

Identify which type of translator would do this.

..... [1]

(iii) One translator creates an error report displaying all the errors in the code before it can be executed.

Identify which type of translator would do this.

..... [1]

(iv) One translator creates an executable file.

Identify which type of translator would do this.

..... [1]

4 An interrupt is a type of signal that is used in a computer.

(a) State the name of the type of software that manages interrupts.

..... [1]

(b) Describe how interrupts are used when a key is pressed on a keyboard.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(c) Interrupts can be hardware based or software based.

A key press is one example of a hardware interrupt.

(i) Give **two** other examples of a hardware interrupt.

1 .....

.....

2 .....

.....

[2]

(ii) Give **two** examples of a software interrupt.

1 .....

.....

2 .....

.....

[2]

5 A student uses both system software and application software on their computer.

(a) Give **one** example of system software.

..... [1]

(b) Give **two** examples of application software.

1 .....

2 .....
[2]

(c) Describe the difference between system software and application software.

.....

.....

.....

.....

..... [2]

6 A computer has an operating system.

(a) The operating system provides several functions.

(i) Tick (✓) **one** box to show which of these is **not** a function of the operating system.

A managing files ☐

B providing an interface ☐

C handling interrupts ☐

D loading the bootstrap ☐

[1]

(ii) Identify **one** other function of an operating system.

Describe the purpose of this function.

.....

.....

.....

..... [2]

(b) Give the name of the set of instructions that are provided to the operating system to allow it to run.

..... [1]

7 Different types of software can be run on a computer.

(a) State what is meant by software.

.....

..... [1]

(b) Utility software is one type of software that can be run on a computer.

Tick (✓) **one** box to show which software is an example of utility software.

A spreadsheet ☐

B anti-virus ☐

C web browser ☐

D database ☐

[1]

(c) Identify the type of software that manages inputs and outputs for the computer.

..... [1]

8 Programs can be written in a high-level language or a low-level language.

(a) Machine code is one type of low-level language.

Identify **one** other type of low-level language.

Identify the translator that is needed for the type of low-level language you have identified.

Type of low-level language .....

Translator .....

[2]

(b) A high-level language can be translated into a low-level language, using a compiler or an interpreter.

Describe the advantages of using an interpreter instead of a compiler during software development.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Integrated development environments (IDEs) provide translators.

Identify **three** other common functions of an IDE.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

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9 A student records himself singing a song for his music project. The recorded sound is converted to binary to be processed by a computer.

(a) The student uses sound editing software to edit the recorded sound.

Tick (✓) **one** box to show whether sound editing software is an example of application, security, system or utility software.

A application software ☐

B security software ☐

C system software ☐

D utility software ☐

[1]

10 A computer has an operating system.

One function of the operating system is handling interrupts.

(a) Describe the role of the operating system in handling interrupts.

.....

.....

.....

..... [2]

(b) Give **two** other functions of an operating system.

1 .....

2 .....

[2]

(c) Hardware and software interrupts are two types of interrupt.

(i) Give **one** example of a hardware interrupt.

..... [1]

(ii) Give **one** example of a software interrupt.

..... [1]

11 A programmer uses assembly language to create a program for a coffee machine.

(a) Describe what is meant by assembly language.

.....

.....

.....

..... [2]

(b) Identify the translator needed for assembly language.

..... [1]

(c) Tick (✓) **one** box to show which statement is an advantage of using assembly language.

A It is close to human language. ☐

B It is memory efficient. ☐

C It is easy to debug. ☐

D It is machine independent. ☐

[1]

12 An integrated development environment (IDE) has two types of translator: a compiler and an interpreter.

(a) Give the purpose of the compiler and the interpreter.

.....

..... [1]

(b) Identify **one** other type of translator.

..... [1]

(c) The table contains statements about compilers and interpreters.

letter	statement
A	An interpreter translates and executes the code one variable at a time.
B	A compiler displays an error report when it is unable to run the code due to syntax errors.
C	An interpreter stops translating as soon as it finds an error in the code.
D	A compiler translates the whole code at once and then it is executed.
E	A compiler produces a non-executable file.

Two of the statements in the table are **not** correct.

Give the letter of each incorrect statement.

Suggest a corrected statement for each.

letter of the incorrect statement 1 .....

corrected statement .....

.....

.....

letter of the incorrect statement 2 .....

corrected statement .....

.....

.....

[4]

(d) A translator is one function of an IDE.

Identify **two** other functions of an IDE.

State the purpose of each function.

function 1 .....

purpose .....

.....

.....

function 2 .....

purpose .....

.....

.....

[4]

13 A programmer uses a low-level language to write a program for a vending machine.

(a) State what is meant by a low-level language.

.....

..... [1]

(b) Identify **two** examples of a low-level language.

1 .....

2 .....
[2]

(c) Give an example of a translator that can be used for a low-level language.

..... [1]

(d) State **two** reasons why the programmer chose a low-level language instead of a high-level language to write the program.

1 .....

.....

2 .....

.....
[2]

(e) The programmer uses an integrated development environment (IDE) to write the program.

Identify **three** terms from the given list that are functions of an IDE.

auto-correction	binary converter
internet protocol (IP) address	media access control (MAC) address
machine independence	prettyprint
protocol identifier	run-time environment

1 .....

2 .....

3 .....
[3]

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END OF QUESTION PAPER